Abdulkader Tarrab Refaee

Engineering Team Lead · Senior Software Engineer · AI / Full-Stack / IoT

*I lead engineering teams and ship full-stack systems end-to-end — from **connected IoT applications and mobile experiences**, through **Flutter / Next.js / Laravel** platforms, to **GPT-4 / Claude / Whisper AI pipelines** in production for newsrooms used by **Al Jazeera** and **Middle East Eye**, and as **technical partner & lead mobile engineer** at **Kantesti** — an AI medical platform with **2M+ users in 127 countries**, partnered with **Microsoft, NVIDIA and Google Cloud**.*

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Open to: Senior / Staff Software Engineering · AI & ML Engineering · Embedded-Systems Leadership · Founding / Principal Engineer roles

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What makes my profile rare

I am the rare combination of an **engineering team lead**, a **senior software engineer**, an **AI engineer**, a **full-stack developer**, and an **IoT engineer** — and I have production credentials in each.

> **Team leadership × AI / LLM engineering × Full-stack on Flutter / Next.js / Laravel × IoT / connected devices × Arabic-first 9-language delivery**

Concrete proof:

- **Al Jazeera- and Middle East Eye-grade** AI media pipelines (via Wisdar Network).
- **Kantesti** — AI medical platform with **2M+ users across 127 countries**, partner of **Microsoft, NVIDIA and Google Cloud**, where I serve as **technical partner & lead mobile engineer**.
- **MWF Metal Detectors** — leading the engineering team behind a global industrial-IoT product line in Istanbul.
- **4 mobile apps live on Google Play**, **50+ websites in production**, an engineering team of **8 + 2** under me.

Professional Summary

Senior software engineer and team lead with **10+ years** of production delivery across **AI/LLM platforms**, **embedded firmware**, **mobile**, and **full-stack web**. Currently:

- **Senior Software Engineer & Embedded Team Lead** at **MWF Metal Detectors** — owning the firmware roadmap on **STM32 (FreeRTOS) / ESP32-C3**, the custom BLE stack, and the flagship **Maxterm** Flutter app (8 detection systems, 42 screens, ~27k LoC, custom 3D engine, 9 languages).
- **AI & Media Systems Engineer** at **Wisdar Network** (clients: **Al Jazeera**, **Middle East Eye**) — building the **Ekteb.ai** SaaS, the `ekteb-ai` Laravel 11 + GPT-4 + SBERT distribution backend, the `ekteb-v2` newsroom platform, and the multi-platform **video_downloader** API.
- **Independent Tech Lead** since 2013 — **40+ commercial products**, **4 apps live on Google Play** (KopKap, Meem Tag, AI Blood Test Analyzer, EDLL Shop), **50+ production websites**, with a team of **8 engineers + 2 designers** under me.

Master's in **AI & IoT** in progress.

At a Glance

- **10+ years** building production software across web, mobile, AI and embedded systems.
- **40+ commercial products** delivered; **4 apps live on Google Play**; **50+ websites in production** across 4 continents.
- Engineering on AI/media platforms used by **Al Jazeera** and **Middle East Eye** (via Wisdar Network).
- **Led a team of 8 engineers + 2 designers**, mentoring, code review, system design, RFCs/ADRs.

- **Technical partner & lead mobile engineer** at **Kantesti** (kantesti.net) — an AI medical platform interpreting **15,000+ blood biomarkers** in under 60 seconds at **99.84% accuracy**, with **2M+ users across 127 countries**, **75+ languages**, **HIPAA + GDPR compliant**, ISO certified, and partnered with **Microsoft, NVIDIA, and Google Cloud**. The companion Flutter app I lead (`AI Blood Test Analyzer`) hit **100k+ downloads on Google Play in two months**.
- Worked on **security-sensitive workflows**: online payments, medical/clinical data, multi-currency escrow.
- Deep hardware-software ownership: **custom BLE protocols** with checksum & 9 concurrent streams, **FreeRTOS on STM32**, **Modbus / CAN / RS-485**, **PCB design (KiCAD, Altium, EasyEDA)**.
- Production AI: **Whisper transcription**, **OpenAI GPT-4o + Claude APIs**, **embeddings + RAG**, **MCP/agents**, **self-hosted Ollama / Llama 3 / Mistral / DeepSeek**.
- Localization at scale: **9 languages with full RTL** in production apps.

Core Strengths

- **Hardware ↔ software end-to-end:** custom BLE / Classic Bluetooth protocols, ESP32-C3, STM32 + FreeRTOS, Arduino, Raspberry Pi, PLC, Modbus / CAN / RS-485, OTA-ready firmware, PCB design.
- **Cross-platform mobile:** Flutter (Riverpod · Bloc · Provider · GoRouter) shipping iOS / Android / Web from a single codebase, with **Fastlane / Codemagic / Bitrise** CI/CD and Play Store / App Store / TestFlight distribution.
- **High-performance web:** Laravel 11/12 + Inertia + React 19 + Vite, Next.js 15 (App Router, Server Components, ISR) with Clean Architecture, Headless / Hybrid WooCommerce.
- **AI/LLM engineering:** GPT-4o + Claude APIs, function calling and structured outputs, Whisper, embeddings, RAG, **MCP / Model Context Protocol**, multi-agent + tool use, prompt engineering, self-hosted open-source models.
- **DevOps & cloud:** Docker (multi-stage, docker-compose), Kubernetes (working knowledge), CI/CD (GitHub Actions, GitLab, Bitbucket, Jenkins), Nginx / Traefik / Caddy, AWS (EC2 / S3 / Lambda / RDS / CloudFront), GCP (Cloud Run / Functions), Cloudflare (Workers, R2, WAF), VPS hardening.
- **Real-time systems:** Agora RTC + RTM, Pusher / Laravel Echo, WebSockets / SSE, gRPC, MQTT for IoT, escrow & negotiation flows.
- **Security-aware delivery:** OAuth 2.0 + OpenID Connect (PKCE, JWT), OWASP Top 10 hardening, TLS configuration, encryption at rest / in transit, WAF / Cloudflare rules, payments and medical workflows.
- **Engineering leadership:** mentoring, code review at scale, RFCs, Architecture Decision Records, system-design diagrams, Agile / Scrum on JIRA, ClickUp, Trello, Git/GitHub workflows.

Professional Experience

Senior Software Engineer & Embedded Team Lead · *MWF Metal Detectors*

2021 – Present · Istanbul, Turkey

- **Lead the embedded-firmware engineering team** that builds the on-device software for MWF's metal/gold detectors — owning the technical roadmap for **microcontroller programming on STM32 (FreeRTOS) and ESP32-C3**, the custom BLE stack, and the cross-stack debugging from MCU → BLE → mobile app.
- Architect and maintain the company's flagship Flutter app **Maxterm** for industrial metal and gold detectors — **8 detection systems, 42 screens, ~27,000 lines of Dart**, custom BLE protocol with checksum and 9 concurrent streams, hand-built 3D engine for ground-scan visualization, 9 languages with full RTL.
- Lead the **global e-commerce platform** `mwf-metaldetectors.com` on WooCommerce — 6 languages, device registration, warranty management, ERP integration, custom shipping plugins.
- Authored and shipped two open-source WooCommerce plugins for Turkish shipping carriers (`basitkargo_woocommerce` , `shipentegra-woocommerce`).
- Designed and deployed the company's CI/CD, container, reverse-proxy and observability setup across staging and production.

AI & Media Systems Engineer · *Wisdar Network — parallel engagement*

2024 – Present · Istanbul, Turkey · Remote/hybrid Wisdar Network is an Istanbul-based media-tech company building AI tooling for major newsrooms — clients include **Al Jazeera** and **Middle East Eye**. As a senior contributing engineer I own multiple platform systems:

- **Ekteb.ai** (landing.ekteb.ai) — AI-powered video production SaaS that automates ~95% of editing, translation (40+ languages), titles, descriptions, covers, subtitles and SEO; integrated with YouTube, TikTok, Instagram and X.
- `ekteb-ai` **distribution backend (Laravel 11)** — the LLM-powered Arabic news/sports distribution engine that pushes content to external platforms (see *Selected Projects → ekteb-ai*).
- `ekteb-v2` **newsroom platform (Laravel 12 + Inertia + React 19)** — the operations dashboard, video pipeline (download → cut → transcribe → translate → montage → publish), and quota/feed engines.
- `video_downloader` **API (Python)** — multi-platform downloader with proxy rotation, browser fingerprinting and S3 delivery.
- `scraper` **article-feed engine (Laravel)** — content-fingerprinting deduplication, prompt registry, distribution rules.

Flutter & Full-Stack Engineer · *International Clinics*

2019 – 2021 · Istanbul, Turkey

- Designed and shipped **GplusClinic** — a Flutter mobile application for International Clinics' medical-tourism operation in Istanbul. Owned the app **end-to-end**: architecture, UI/UX implementation, state management, API integration with the Laravel/WordPress backend, store deployment and post-launch support.
- **Multilingual** delivery for the medical-tourism market (Arabic with full RTL, English, Turkish), surfacing treatments, clinics, doctors and care packages tailored to international patients.
- Patient-facing flows: **consultation requests**, **appointment booking**, **medical-history and document upload**, **multi-currency price display**, **push notifications** for appointment reminders and clinic communication.
- Backend integration with the clinic's CRM and case-management workflow so clinic staff could service inbound mobile leads in the same pipeline as web leads.
- Delivered the multilingual flagship site `internationalclinics.com` on **Laravel + WordPress** with a strong visual identity and an SEO foundation tuned for international medical-tourism keywords.

Independent Software Engineer & Tech Lead *(consulting practice)*

2013 – Present · Turkey / remote A long-running independent practice running in parallel with full-time roles, covering full-stack web, mobile, AI/LLM pipelines and embedded systems for clients across Turkey, the GCC, Europe, North America and South-East Asia.

- **Engineering leadership:** led a team of **8 software engineers + 2 designers** on multi-app deliveries (Laravel APIs + 3+ Flutter apps in parallel), with code reviews, RFCs and Architecture Decision Records.
- **Technical partner & lead mobile engineer** at **Kantesti** *(minority equity)* — AI medical platform (Cologne, DE / London, UK · founded 2019) interpreting blood-test results across **15,000+ biomarkers**, **2M+ users in 127 countries**, **75+ languages including Arabic**, **HIPAA + GDPR compliant**, partnered with **Microsoft / NVIDIA / Google Cloud**. The Flutter companion app I own end-to-end (`AI Blood Test Analyzer`) hit **100k+ Google Play downloads in two months** — my largest mobile deployment to date.
- Built **AI/LLM-driven content pipelines** (Ekteb / video_downloader): Whisper transcription, OpenAI + Claude orchestration, embeddings, RAG, MCP-based agents, anti-ban scraping with proxy rotation and browser fingerprinting.
- Shipped **4 mobile apps live on Google Play** (KopKap, Meem Tag, AI Blood Test Analyzer, EDLL Shop) and 50+ websites — see *Published Mobile Apps* and *Production WordPress & WooCommerce Portfolio*.

Education

- **Master's Degree, Artificial Intelligence & IoT** — Turkey, 2020 – Ongoing
- **B.Sc., Electrical & Electronic Engineering** — University of Aleppo, Syria, 2011 – 2018 (70%).
- **Diploma, Computer Maintenance** — Fifth Institute of Industry, Aleppo, 2007 – 2011 (97%).

Published Mobile Apps (Google Play)

App	Package	Role	Stack
KopKap — Pet marketplace (live streaming, escrow, multi-currency)	`com.kopkap.app`	Lead developer (mobile + backend)	Flutter · Riverpod · Laravel 11 · Stripe · Agora · Firebase
Meem Tag — Multi-vendor marketplace (Matajiri network)	`com.matajiri.meemtag`	Mobile developer	Flutter · Laravel modular backend · Firebase
AI Blood Test Analyzer *(Kantesti · 100k+ downloads in 2 months · 75+ languages)*	`com.aibloodtestanalyzer.app`	Technical partner & lead mobile engineer	Flutter · `flutter_inappwebview` · Dio · multilingual onboarding
EDLL Shop — Two-sided e-commerce app (Matajiri network)	`com.matajiri.Edllshop`	Mobile developer	Flutter · companion User & Seller apps · Laravel API

> Earlier published apps include the **GplusClinic** medical-tourism app (International Clinics, Flutter) and the **Maxterm** companion app for MWF metal detectors (private distribution to dealers).

Selected Projects (with the engineering ideas worth highlighting)

> 46 repositories across `abdulkaderTarrabRefaee` , `Wisdar-Network` , `theengineer00000` , `wisdarnetwork` and `A-Hadi-Altallawi` on GitHub.

1. Maxterm — MWF Detectors flagship app *(Flutter · BLE · 3D · 9 languages)*

A Flutter application for an industrial metal/gold detector that ships **eight independent detection systems** (GEO, LDS, Ground Scan, Ground Sensor, Hybrid, ION, Live Scan, U-VLF), 42 screens and ~27,000 lines of Dart. **Engineering highlights**

- **Custom BLE protocol** over `flutter_blue_plus` with checksum validation and **9 concurrent data streams** between phone and STM32 / ESP32-C3 firmware.
- **Hand-rotated, hand-projected 3D engine** (no third-party 3D library) for the Ground Scan terrain view — interactive rotation, zoom and wireframe/grid overlays driven by `vector_math`.
- Live target visualization with `fl_chart`, `flutter_echarts`, `flutter_map_heatmap`, animated gauges, percent indicators and Lottie.
- Sensor fusion: `flutter_compass`, `pedometer`, `geolocator`, `vibration`.
- Localization in **9 languages including RTL** (Arabic, Persian, English, Turkish, Malay, German, Spanish, French, Russian).
- State management with **Provider**; persisted scan history via `shared_preferences`.

2. Kopkap — Pet Marketplace *(Flutter app + Laravel 11 API)*

A global marketplace for pets and certified breeders, available on iOS, Android and the web. **Engineering highlights**

- **Backend (Laravel 11):** Sanctum auth, Spatie Translatable + Query Builder, Firebase (Cloud Messaging + Auth), **Stripe Connect with escrow** (funds held until buyer confirms), Pusher real-time, AWS S3 + CloudFront, Intervention Image, JWT, PHPMailer.
- **Mobile (Flutter): Riverpod** state, **GoRouter** navigation, Dio + Connectivity, Firebase Messaging + Firestore + App Check, Google/Apple Sign-In, **Agora live streaming**, video reels, **flutter_stripe**, secure storage, deep links, geocoding, Google Maps, photo viewer.
- **Domain features**: multi-currency wallet with automatic FX, structured **negotiation/counter-offers**, refunds with **independent dispute arbitration**, geo-search by radius, multi-language UX (Arabic RTL default + English).

3. Ekteb — AI Content & News Platform *(Laravel 12 + Inertia + React 19 + TypeScript · team delivery)*

Newsroom/automation platform built for **Wisdar Network** (clients: Al Jazeera, Middle East Eye) that **downloads videos → transcribes → translates → montages → posts to social media**. *(Built with a team — I am one of the engineers contributing to the platform, not the sole author.)* **Engineering highlights**

- Modern Laravel 12 + **Inertia + React 19** + Vite + Tailwind v4, with **shadcn/Radix UI** primitives and i18next for i18n.
- 9-state video pipeline (`downloading → cutting → analyzing → readyForReview → montaging → quality_control → ready_for_publishing → delivered → postedOnSocialMedia`) plus dedicated failure states.
- **Article-feed engine with content fingerprinting / deduplication**, prompt management dashboard, statistics with quota resets, Meilisearch-style search, Excel & Word import/export (Maatwebsite, PHPWord).
- Real-time presence and notifications via **Agora RTM 2.x**, Pusher and Laravel Echo.
- AWS S3 + Glacier flows, AI prompt registry per platform/language.

3b. ekteb-ai — LLM-Powered Distribution Backend *(Laravel 11 · GPT-4 · SBERT · Azure MySQL)*

Companion backend to Ekteb (also for **Wisdar Network**): a Laravel 11 platform that distributes Arabic sports/news content to external publishers with intelligent LLM-driven filtering and matching. **Engineering highlights**

- **Service-Oriented Architecture** with clean separation: `Controllers → Services → Models`.
- **8 polymorphic content types** (`Cover`, `Short`, `Article`, `Sub`, `Cut`, `Feed`, `ArticleFeed`, `VoiceOver`), each with its own dedicated table and lifecycle.
- **Smart Distribution Engine** with 3 routing modes: `balanced`, `fair`, `bestMatch` — picks the optimal publisher per content piece.
- **Quota Management**: per-platform daily/hourly publishing quotas with automatic resets.
- **LLM-Powered Filtering**: customizable GPT-4 prompts per platform/category for classification, relevance and editorial rules.
- **Cross-lingual deduplication** using **Python SBERT** for semantic similarity (branch: `feature/cross-lingual-deduplication`).
- **Multi-database** topology: local app DB + external **Azure MySQL** crawler DB.
- Full **activity logging** for every operation (audit-grade traceability).
- Stack: Laravel 11 · PHP · Eloquent ORM · MySQL · OpenAI GPT-4 · Python SBERT · Blade · Laravel Pint.

4. Headless Hybrid WooCommerce Storefront *(Next.js 15 · React 19 · Clean Architecture)*

Replaces the front-end of an existing thousand-product WooCommerce store with a Next.js 15 storefront — keeping cart and checkout on WordPress. **Engineering highlights**

- **Strict Clean Architecture** with 4 layers (`domain/`, `application/`, `infrastructure/`, presentation); enforced **inward-only dependency** rule and mandatory mappers between GraphQL types and domain entities.
- WPGraphQL + WPML integration where the Next.js app **only knows about a `locale` variable** — WPML never leaks into domain.
- **Meilisearch + react-instantsearch + InstantSearch.js** for sub-100 ms search, **Zustand** for cart state.
- ISR + on-demand revalidation tagged per `product:slug`; **never `generateStaticParams` for all products** (top 200–500 only).
- Performance targets met: **LCP < 1.5s**, **TTFB < 200ms**, **Lighthouse 95+**, vs. 50 baseline on the legacy theme.
- Docker + Nginx reverse proxy on a single domain, TypeScript strict, Tailwind v4, pnpm.

5. Video Download API with Anti-Ban Arsenal *(Python · S3)*

A REST service for **Wisdar Network** that downloads videos from any platform → uploads to S3 → returns a presigned URL. **Engineering highlights**

- **Proxy rotation** with three strategies (Round Robin / Random / Least-Failed) over HTTP/HTTPS, SOCKS5, and rotating providers (Bright Data, Oxylabs, SmartProxy, Webshare, Decodo). Failing proxies are auto-banned after 3 failures.
- **Browser fingerprinting**: matched `User-Agent` ↔ `Sec-CH-UA` ↔ `Sec-CH-UA-Platform`, randomized `Accept-Language` and `Sec-Fetch-*` — any mismatch instantly flags a bot.
- **Cookie rotation** per platform (`youtube.txt`, `instagram.txt`, …), browser-extracted cookies (`BROWSER_COOKIES=chrome`).
- **PO-token generator** for YouTube hardening, Facebook-specific cookie fix, high-quality trim pipeline.
- Dockerized with Apache deployment; protected admin dashboard with auth (`init_auth_db.py`).

6. EuroMarketn — B2B Wholesale Platform for Europe *(Laravel + 2 Flutter apps)*

euromarketn.com · A full B2B wholesale platform with **separate apps for merchants and end-users**, built to serve the European market end-to-end.

Engineering highlights

- **Backend (Laravel + PHP):** custom catalog with multi-currency pricing, bulk-order flows, supplier-onboarding pipeline, role-based dashboards for merchants, customers and admins; **Blade + Bootstrap + JavaScript** for the web admin and storefront.
- **Two Flutter apps shipping in parallel:**
  - `EuromarketnUserApp` — buyer-facing app with bulk-order quoting, cart-to-quote flow, order tracking and Firebase push notifications.
  - `EuroSeller` — merchant-facing app with inventory, order management, fulfillment, pricing and reporting.
- A separate `eurobas_car` module for an automotive-parts vertical inside the same ecosystem.
- **Firebase** notifications + auth, multi-language storefront targeting European markets.
- Full ownership: backend, both mobile apps, store deployment, and the merchant onboarding workflow.

7. Meemtag — Multi-Vendor Marketplace *(Laravel modular + 2 Flutter apps live on Google Play + Next ERP integration)*

Companion app **Meem Tag** on Google Play (developer: **MATAJIRI** network).

A multi-vendor marketplace with a **modular Laravel** architecture designed to be reused as a white-label SaaS — sellers, products, orders, payments, shipping, ratings, a full installation wizard, **and a custom programmatic bridge to Next ERP**.

Engineering highlights

- **Modular monolith** architecture: each business domain (vendors, catalog, orders, payments, shipping, ratings, CMS) lives in an independent module via `modules_statuses.json` — modules can be enabled/disabled per tenant.
- **Self-hosted installation wizard** for white-label deployment by partners (database setup, environment config, admin bootstrap).
- **Two Flutter apps:** `MeemtagUserApp` (buyer) and `MeemtagSellerApp` (vendor) — both live, with the user app on Google Play under the Matajiri brand.
- **Multilingual storefront** (Arabic / English / Turkish at minimum) with full RTL.
- Vendor onboarding, KYC, payouts, shipping integrations and product moderation flows.
- Ships as the technical backbone of the broader **Matajiri** marketplace network (see also EDLL Shop, `com.matajiri.Edllshop`).
- **Next ERP integration (custom-built):** a programmatic bridge I designed and shipped between the Meemtag Laravel backend and **Next ERP**. Bi-directional sync of products, stock levels, customers, orders, invoices and accounting entries. Built on REST APIs, signed webhooks for real-time events, idempotent job queues for retry-safety, and field-mapping layers so the marketplace catalog and the ERP master-data stay consistent. The integration lets the marketplace operate as the storefront/order entry while accounting, inventory and finance live in Next ERP — a clean Headless commerce ↔ ERP split.

8. ServiceQo — On-demand Service Marketplace *(Laravel 8 + 3 Flutter apps)*

A full handyman/booking marketplace platform: customers book services, providers fulfill them, and an admin dashboard manages the entire ecosystem. **Three Flutter apps in production** — User, Provider, and an internal/admin app — backed by a single Laravel API.

Engineering highlights

- **Backend (Laravel 8 + PHP 8):** Sanctum auth, **Spatie Permission** (RBAC) and **Spatie MediaLibrary**, **DomPDF + mPDF** for invoices and contracts, **Yajra DataTables** for the admin dashboard, **Laravel Geocoder** for location-aware booking, **AWS S3 + Flysystem** for media, **Ziggy** for PHP→JS routes, OneSignal notifications, Stripe payments.
- **Mobile (Flutter, MobX state):** three Flutter apps — `booking_system_flutter` (user), `handyman_provider_flutter` (provider) and `serviceqo_app` (internal). Each integrates the Firebase suite (Auth, Crashlytics, Storage, Firestore, Remote Config, Cloud Messaging, Pagination), local notifications, Apple/Google Sign-In, image and file pickers, Lottie, Custom Tabs and WebView.
- **Multi-gateway payments on the provider app** (international coverage in a single codebase): **Stripe, Razorpay, PhonePe, Flutterwave, CinetPay** (West Africa), **PayPal Checkout, midpay**.
- Provider app also ships **Syncfusion charts** for earnings analytics, **Geocoding** for service-area mapping, and a **OneSignal + FCM hybrid** push.

9. WooCommerce Plugins (PHP, public)

- `basitkargo_woocommerce` — fixes broken official Basit Kargo integration; advanced API calls, logging, debugging.
- `shipentegra-woocommerce` — official ShipEntegra integration, API v2.6.29, label generation, dual sandbox/production, comprehensive logging.

10. ArduCar — Bluetooth RC Car *(Flutter + Arduino, public)*

- Phone-as-controller talking to **HC-05 SPP** module on Arduino Uno.
- **Two control modes**: tilt (accelerometer) and on-screen buttons with **exponential speed ramping (PWM 60→255 over 2 s)**.
- ASCII protocol `X,Y,Z\n` (steering / speed / reserved) at **10 Hz / 9600 baud**.
- Differential-drive over **L298N** dual H-bridge; **600 ms safety timeout** kills motors on signal loss.

11. Kantesti — AI Medical Platform · *(technical partner & lead mobile engineer · minority equity)*

kantesti.net · founded 2019 · HQ in Cologne, Germany · offices in London, UK Companion app **AI Blood Test Analyzer** on Google Play

A globally-deployed AI medical platform that interprets blood-test results from a photo or PDF in under 60 seconds — across **15,000+ biomarkers** at a stated **99.84% accuracy**. I serve as a **technical partner and lead mobile engineer** (minority equity holder, not founder) — owning the Flutter companion app end-to-end and contributing to the broader technical architecture.

Scale & credibility

- **2,000,000+ users** across **127 countries**.
- **100,000+ Google Play downloads in the first two months** of the companion Flutter app.
- **75+ languages** supported, including Arabic.
- **HIPAA + GDPR compliant**, multiple **ISO certifications**.
- **Strategic partners: Microsoft · NVIDIA · Google Cloud**.

Product surface

- Comprehensive analysis: complete blood count (CBC), liver / kidney function, lipid panel, thyroid, hormones, vitamins.
- Personalized **7-day meal plans**, supplement recommendations, longitudinal trend comparisons across past tests.
- Family-level **hereditary health-risk assessment**.
- Tiered pricing (free single report, paid single report ~₺267.99, annual subscription ~₺1499 for 12 reports + premium features), enterprise/B2B offering.

My contribution

- **Flutter companion app, end-to-end** (`com.aibloodtestanalyzer.app`) — `flutter_inappwebview` , Dio HTTP, fine-grained permissions, multilingual onboarding, secure storage, App Store / Google Play deployment.
- Engineering input on the multilingual web platform and **payment processing** flows.
- **Security-sensitive medical workflows** — privacy by design, encrypted transit, payment-grade hardening, alignment with HIPAA / GDPR.

12. EdllShop — User & Seller apps *(Flutter, public)*

Two-sided e-commerce (`EdllShopUserApp` , `EdllShopSellerApp`) on Flutter.

13. Earlier portfolio (production)

- `mwf-metaldetectors.com` — global WooCommerce store (6 languages, ERP-integrated).
- `kudokudo.com` — global restaurant chain (WordPress).
- `safarna.com.tr` — Istanbul tourism site (WordPress) + travel management (`pro.safarna.com.tr`) + reviews (`feed.safarna.com.tr`) on Laravel/PHP.
- `internationalclinics.com` — multilingual medical-tourism site.

14. Autonomous Vision-Guided Robot Car *(graduation project, 2018)*

A Raspberry-Pi-based mobile robot built as my B.Sc. graduation project — an early proof-of-concept that combined embedded compute, computer vision and remote control into a single platform.

Engineering highlights

- **Compute & I/O:** Raspberry Pi 3 with a camera mounted on **two servo motors** for **full 360° pan-tilt movement** under software control.
- **Computer-vision pipelines (Python + OpenCV):** real-time **hand recognition and tracking**, **ball detection and tracking** (color and shape segmentation), object-following control loop.
- **Live video streaming** from the on-robot camera to a remote operator.
- **Remote control** layer over the network — drive, steer the camera, and switch between autonomous-tracking and manual modes.
- Fully integrated firmware → vision → networking → control stack on a single embedded Linux device.

This project anchors my long-running interest in computer vision and IoT/edge devices that has continued throughout my career.

Tools I Built

Internal/utility tools I designed and shipped to make my own (and my teams') work faster — they are utilities, not commercial products, and are listed separately so they don't compete with full platforms above.

Trendyol → WooCommerce Importer *(Python · Playwright · PostgreSQL · Redis)*

Smart crawler against Trendyol with AI-powered translation and content rewriting, rule-driven auto-pricing, and bi-directional price/stock sync. PostgreSQL for state, Redis for caching, a REST middleware that pushes to WooCommerce.

WooCommerce Bulk Uploader v4 *(Python)*

Bulk product uploader from local folders into WooCommerce — `.env` credentials, retry logic, **SKU-based deduplication**, image compression, progress bar, **resume after disconnection**, dry-run mode, HTML report, Telegram notifications.

Bluetooth mwfTerminal *(Java / Android · public)*

Bluetooth Serial Port Profile (RFCOMM, UUID `00001101-...`) terminal used to **debug detection-device firmware** on classic Bluetooth — internal diagnostic tool used during MWF firmware development.

Translate-lang for Laravel *(Python)*

Auto-translates Laravel `lang/` files between languages via Python — used to roll out new languages quickly across the WordPress / Laravel portfolio.

Echarts 3D Surface for Flutter *(Dart · public)*

A Flutter widget that bridges **Apache ECharts 3D** into Flutter via WebView, with two-way communication between Dart and JavaScript and reactive updates whenever the chart options change. Used as a building block in detector visualization and dashboard charts.

Production WordPress & WooCommerce Portfolio (selected)

A long-running practice covering corporate sites, e-commerce, multilingual content platforms and industry-specific landing pages — built on **WordPress · WooCommerce · Elementor · custom themes/plugins**, with multilingual setups (Arabic, English, Turkish, Russian, German, Filipino…), payment & shipping integrations, SEO and analytics.

E-commerce & Retail (WooCommerce)

Project	Domain	Sector
MWF Metal Detectors (US)	mwf-metaldetectors.com	Industrial detectors — 6 languages, ERP, warranty system
Vogel Detectors (Germany)	vogeldetectors.com	Gold & metal detectors
MWF PH (Philippines)	mwf-ph.com	Metal detectors – PH market
Sensarix / Syria Detectors	syria-metaldetectors.com	Metal detectors – Syrian market
Türkiye Store	ty-store.net	Generalist Turkey marketplace
Osmanli Shop	osmanlishop.com	Ottoman heritage products
Oman-Turk Group	omanturkgroup.com	Omani-Turkish electronics
Mech Store	mech-store.com	Mechanical parts
LED Manya	ledmanya.com	Turkish LED & electronics
İstediğin Her Şey Burada	istediginherseyburada.com	Turkish marketplace
Mirabella Professional	mirabellaprofessional.com	Beauty/professional supplies
Seyidoğlu Sweets	shop.seyidoglu.com	Famous Turkish sweets brand
Kırtas Shop	kirtas-shop.com	Stationery & office supplies
Matajiri	matajiri-tr.com	Multi-vendor Turkey
Charco Chicken	charcochicken.com.tr	Healthy-chicken food chain
LOOZ Nuts (Germany)	looznuts.de	Premium roasted nuts, dates, nut butters — Munich, DE/EN
Pellabez (Turkey)	pellabez.com	Household, kitchen & cleaning, B2B-ready, TR/EN
Shoppy Pella	shoppypella.com	E-commerce sister brand to Pellabez
Istinye Turkish Sweets (UAE)	istinyeturkishsweets.com	Authentic Turkish sweets shop — Dubai
KaliKsa (Turkey)	kaliksa.com	Custom-printed apparel — Kocaeli

Hospitality, Restaurants & F&B

- **Kudo Kudo** — global fried-chicken franchise (50+ locations worldwide), main site + Turkey + Kenya branches · kudokudo.com · kudokudo.com.tr · kenya.kudokudo.com
- **Sultan's Grill** — Sultan restaurant in Canada · sultansgrill.ca
- **Charco Chicken** — healthy-chicken food chain · charcochicken.com.tr
- **Istinye Turkish Sweets** — Turkish confectionery store, Dubai · istinyeturkishsweets.com
- **LOOZ Nuts** — German premium nuts brand, Munich · looznuts.de

Real Estate, Architecture & Interior

- **Kudus Group** — Jerusalem Real Estate in Turkey · kudusgroup.net
- **Tactic Group** — tourism / real estate / medical · taktiktravel.com
- **Habbash Group** — interior design · habbashgroup.com
- **Elva Architecture** · elva-architecture.com
- **Al Khayyal Umbrellas** (Saudi Arabia) · alkhayyals.com
- **Bayan Al Madina** — engineering consultancy, Saudi Arabia (Medina) · bayanmedina.com
- **Mazen Curtains** — custom curtains & drapery, Surrey, BC, Canada · mazen-curtains.com
- **Modern Homes** — Dubai real-estate agency (Emaar, Azizi, Sobha, Damac partnerships) · modernhomes.ae
- **Mirna DXB** — Dubai luxury property investment platform · mirnadxb.com

Healthcare & Medical Tourism

- **International Clinics** · internationalclinics.com
- **MedTürkiye** — Turkey medical treatment · medturkiye.com
- **SuperEye Laser** — eye surgery · supereyelaser.com
- **Medical White Clinic** · medical-white.net
- **Jawaher Dental** · jawaherdental.com
- **Vitalora Medical** — hair transplant, dental, bariatric, ophthalmology, Istanbul · vitalora.net

Tourism & Education

- **Turlands** — Turkey tourism · turlands.com
- **Study Smart** — student services in Turkey · studysmartofficial.com

Logistics & Services

- **Ayla** — shipping & export · ayla.com.tr

Legal, Editorial & Non-profit

- **Middle East Legal** — UAE legal consultancy · middleastlegal.com
- **Idhaat Station** — blog for Islamic writer Nabil Shabib · idhaat-station.info
- **Waqf Yaffa** — Islamic endowment for heritage preservation, Jaffa · waqfyaffa.com

Creative Agencies & Studios

- **Inspirers Media** — Saudi digital marketing agency (photography · graphic design · motion · web · social), Medina, since 2016 · inspirers-media.com
- **Trika Studio** — photography & videography studio, Saudi Arabia · trikastudio.net

Landing Pages

- **Livin Istanbul** (Real Estate) — `en.livin.ist`, `ar.livin.ist`, `ru.livin.ist`
- **N1CE1** — products landing · n1ce1.online
- **Lennox Clinic — Hair Transplant**: ar-hair.lennoxclinic.com, en-hair.lennoxclinic.com
- **Lennox Clinic — Dental**: ar-dental.lennoxclinic.com, `en-dental.lennoxclinic.com`
- **Venusto Clinic — Cosmetic** · cosmetic.venustoclinic.com

> Across these projects: custom themes & child themes, Elementor / WPBakery, **WooCommerce** with custom payment & shipping plugins (Basit Kargo, ShipEntegra, Stripe, iyzico, local Turkish carriers), **WPML / Polylang** multilingual setups, advanced SEO, performance optimization (caching, image optimization, lazy loading), Google Tag Manager / GA4, Facebook Pixel, REST/GraphQL integration where needed.

Technical Skills

Programming Languages

Dart · TypeScript / JavaScript · PHP · Python · Java · C# · C++ · C · SQL · Bash · Assembly · Prolog

AI / LLM Engineering

- **LLM APIs:** OpenAI (GPT-4o, function calling, structured outputs) · Anthropic Claude (4.x family) · self-hosted **Ollama**, **Llama 3**, **Mistral**, **DeepSeek**.
- **Speech & content: Whisper** transcription pipelines (production use in Ekteb).
- **Retrieval: Embeddings** (text-embedding-3, sentence-transformers) · **RAG** (Retrieval-Augmented Generation, led a team-built implementation) · **Vector databases** (pgvector, Qdrant, Weaviate, ChromaDB — working knowledge).
- **Agents & tooling:** Multi-agent orchestration · **Tool use** · **MCP (Model Context Protocol)** · prompt engineering · lightweight fine-tuning.

Computer Vision & On-device ML

TensorFlow · **PyTorch** · **YOLO** · **TensorFlow Lite / ML Kit** on mobile · **OpenCV** (object tracking, hand recognition, ball tracking, video streaming, remote control)

Cloud

- **AWS** — EC2, S3 + CloudFront, Lambda, RDS, ECS, CloudWatch, Route 53.
- **Google Cloud** — Compute, Cloud Run, Cloud Functions, Firebase suite.
- **Cloudflare** — Workers, R2, Pages, DNS, CDN, WAF rules.
- **Azure** — working knowledge.
- Self-managed VPS at scale (Hetzner, Contabo, DigitalOcean, shared hosts).

DevOps & Reliability

Docker (multi-stage builds, docker-compose, production hardening) · **Kubernetes** (Deployments / Services / Ingress / Helm — working knowledge) · **CI/CD** with GitHub Actions, GitLab CI, Bitbucket Pipelines, Jenkins · Reverse proxies (**Nginx**, **Traefik**, **Caddy**) · Apache · application profiling, N+1 query elimination, multi-layer caching strategies.

Security

OAuth 2.0 / OpenID Connect (PKCE, JWT) · **OWASP Top 10** hardening · **Penetration testing** mindset · TLS configuration · encryption at rest and in transit (AES, RSA) · **WAF / Cloudflare security rules** · payment-grade security (Stripe Connect, escrow flows) · medical-data handling.

Backend

Laravel 11 / 12 (Sanctum · Spatie · Eloquent · Queues · Broadcasting · Inertia SSR) · **Node.js** · **Python** (FastAPI · Flask) · **WordPress + WooCommerce** custom plugins · **WPGraphQL** · **ASP.NET Core** · API design (REST, **GraphQL** with Apollo, **gRPC**, **WebSockets / SSE**) · **Microservices** and **Modular Monolith** patterns · event-driven (Redis Streams, queues, broadcasting; working knowledge of Kafka / RabbitMQ / SQS).

Frontend

Next.js 15 (App Router · Server Components · ISR · on-demand revalidation) · **React 19** · **TypeScript strict** · **Tailwind v4** · **Radix UI / shadcn** · **Inertia.js** · **Vite** · ASP.NET Core Razor · Blade · HTML/CSS · Bootstrap · UI/UX with **Figma**.

Mobile

Flutter (Riverpod · Bloc · Provider · GoRouter · Dio · Hive · flutter_secure_storage) · **Android Studio (Java)** · iOS deployment · **Mobile DevOps:** Fastlane, Codemagic, Bitrise, **Firebase App Distribution**, **TestFlight** · 4 apps live on **Google Play**.

Databases & Search

PostgreSQL (JSONB · CTEs · indexing · partitioning · advanced query tuning) · **MySQL / MariaDB** · **MongoDB** · **Elasticsearch** · **Redis** (caching · queues · rate limiting · pub/sub) · **Hive** (Flutter) · **Firebase Firestore** · production-grade **migrations & schema versioning** · **Meilisearch** instant search.

Real-time, Messaging & Communication

Agora RTC + RTM (live streaming, Reels) · Pusher · Laravel Echo · Socket.IO · WebSockets / Server-Sent Events · **gRPC streaming** · **MQTT** (Mosquitto, HiveMQ) for IoT · Firebase Cloud Messaging · Awesome / Local Notifications.

Embedded Systems & IoT

- **MCUs:** ESP32-C3 · STM32 (with **FreeRTOS**) · Arduino · Raspberry Pi · PLC.
- **Industrial buses:** Modbus · CAN bus · RS-485.
- **Wireless:** BLE · Classic Bluetooth (SPP) · MQTT · LoRaWAN · Zigbee · Wi-Fi mesh.
- **PCB Design:** KiCAD · Altium · EasyEDA.

- Custom binary / ASCII serial protocols with checksums · L298N H-bridges · HC-05 · Servo / DC motor control · Sensor fusion (compass · accelerometer · GPS · pedometer).

Payments & Commerce

Stripe (Connect, escrow, refunds) · **Razorpay** · **In-App Purchase** · multi-currency wallets · structured negotiation / offer flows · dispute arbitration · WooCommerce custom payment & shipping plugins · ERP integrations.

ERP / CRM Development & Integration

Hands-on developer (not just admin) on the major business platforms used in the region, with custom code, module development and end-to-end integrations:

- **Odoo** — custom module development, model extensions, server actions, scheduled actions, custom reports and dashboards, portal/website development, ORM customizations.
- **Bitrix24** — custom application development, REST API integrations, business-process (workflow) automation, CRM and deal-pipeline customizations.
- **Zoho** — Zoho Creator scripting in **Deluge**, Zoho CRM customization, custom modules, workflow rules and blueprint automations.
- **Integration patterns I deploy regularly:**
  - **WhatsApp Business API** ↔ CRM (auto lead capture, conversation logging, broadcast).
  - **Website forms** ↔ CRM (real-time webhook ingestion, field mapping, dedup).
  - **Mobile apps** ↔ CRM/ERP (Flutter ↔ Odoo / Bitrix24 / Zoho via REST + OAuth).
  - **Google Sheets** ↔ ERP (bi-directional sync via Apps Script and webhooks).
  - **WooCommerce / Shopify** ↔ Odoo / Zoho (orders, inventory, customers).
  - **Payment gateways** ↔ CRM (auto-invoice, reconciliation).
  - **Custom REST APIs** for external partners and white-labelled clients.

Testing & Quality Engineering

Unit / Integration tests — **PHPUnit · Pest · Jest · Vitest · Flutter test** · **E2E tests — Playwright · Cypress** · **TDD** discipline · rigorous **code review and PR workflows**.

Architecture & Engineering Practice

Clean Architecture · **Modular Monolith** · Headless / Hybrid CMS · Domain-driven type design (branded types) · Mappers · **RFCs** · **Architecture Decision Records (ADRs)** · system-design diagrams and review · technical writing.

Methodology & Engineering Practice

Agile · Scrum · sprint planning · **TDD** · mentoring & code review for distributed teams · **RFCs** · **Architecture Decision Records (ADRs)** · open-source contributions.

Team Tools & Collaboration

- **Project & sprint management:** JIRA · Trello · ClickUp · **Linear** · **Monday.com** · GitHub Projects · GitLab Issues & Boards.
- **Code hosting:** GitHub · **GitLab** · **Bitbucket** · **Azure DevOps Repos**.
- **Code review & quality automation:** Pull-request workflow · GitHub Code Review · **CodeClimate** · **SonarQube** · **Conventional Commits** · **Semantic Versioning** (SemVer).
- **Communication: Slack** · **Microsoft Teams** · **Discord** · **Telegram** · **Zoom** · **Google Meet**.
- **Documentation & knowledge management: Confluence** (Atlassian) · **Google Workspace** / Google Docs.
- **Diagramming & whiteboarding: Miro** · **Lucidchart** · **Draw.io** · Figma.
- **Time tracking & productivity: Toggl**.
- **CRM & client operations** (for tech-lead / CTO engagements): **Intercom** · **Zendesk** · **HubSpot** · **Zoho** · **Bitrix24**.

Languages

- **Arabic** — Native
- **English** — Fluent
- **Turkish** — Advanced
- **German** — A1

Certifications

- **Android Studio Java + Firebase** — Projectsy Institute

Volunteering

- **Volunteer Teacher** — Coding workshops for underprivileged students in Aleppo, Syria (basic programming and web development).
- **UNDP Volunteer** — Organized well-digging projects in Aleppo, Syria.

References

Available on request.